JAMES MITCHELL

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PROFESSIONAL SUMMARY

Accomplished Research Engineer with 6 years of experience driving innovation in machine learning infrastructure and distributed systems. Proven track record of designing and deploying large-scale research platforms that have enabled cross-functional teams to accelerate development cycles. Passionate about translating theoretical research into production-ready systems and mentoring emerging talent in the field of computational research.

WORK EXPERIENCE

Principal Research Engineer

DeepScale AI (Previously known as Horizon Analytics)

Edinburgh, United Kingdom

January 2022 - Present

Architected and led the development of a distributed machine learning platform serving 45+ research scientists across 8 teams, overseeing a technical budget of GBP 2.3M annually

Designed novel approaches to model training optimization, resulting in 60% reduction in computational costs and enabling real-time inference capabilities for large language models

Directed strategic partnerships with 3 major academic institutions, establishing research collaboration frameworks and publishing 12 peer-reviewed papers in top-tier venues

Managed hiring and onboarding of 18 junior and mid-level engineers, establishing best practices for code review, testing, and continuous integration across the research division

Intermediate Research Engineer

Quantum Dynamics Solutions

https://www.quantumdynamics.io

Cambridge, United Kingdom

September 2020 - December 2021

Developed Python-based simulation frameworks for quantum computing applications, processing datasets exceeding 500GB in batch operations

Implemented containerized microservices architecture using Kubernetes and Docker, reducing deployment time by 45% across development environments

Collaborated with physics research teams to optimize numerical algorithms, achieving 3x performance improvements in spectral analysis computations

Maintained comprehensive technical documentation and delivered 6 technical presentations at industry conferences

Junior Research Engineer

TechVision Research Labs

https://www.techvisionresearch.net

Edinburgh, United Kingdom

June 2018 - August 2020

Contributed to the development of data processing pipelines for computer vision research using TensorFlow and OpenCV

Assisted in designing experiments for image classification models, achieving 94% accuracy on benchmark datasets

Supported infrastructure improvements and system maintenance for the research computing cluster

Participated in code reviews and collaborated with 3 senior researchers on experimental design methodologies

EDUCATION

PhD in Computer Science (Machine Learning)

University of Edinburgh

Edinburgh, United Kingdom

2015 - 2018

Bachelor of Science in Computer Science

University of Manchester

Manchester, United Kingdom

2011 - 2015

TECHNICAL SKILLS

Programming Languages: Python, C++, Java, Go, SQL

Machine Learning Frameworks: TensorFlow, PyTorch, JAX, Scikit-learn

Data Technologies: Apache Spark, Hadoop, PostgreSQL, MongoDB, Redis

Infrastructure and DevOps: Kubernetes, Docker, AWS, GCP, Terraform

Research Tools: Jupyter, Git, LaTeX, Weights and Biases

Specializations: Distributed Systems, Deep Learning, Computer Vision, Data Pipeline Architecture